**DEPARTMENT OF WORKS & HIGHWAYS**

**MATERIAL SCIENCE & TECHNOLOGY BRANCH**

**Materials Testing Laboratory**

☑ **Port Moresby,** ☐ **Lae,** ☐ **Mt. Hagen,** ☐ **Annex Lab.**

Date:

Page of

# CONE LIQUID LIMIT & LINEAR SHRINKAGE

## Procedure: AS1289.3.9, 3.3.1, 3.3.2, 3.4.1, 2.1.1 & 2.1.4

Client : ----------

Project: ----------

Location: ----------

Material Description: ----------

Asset No.: ----------

Material Source : ----------

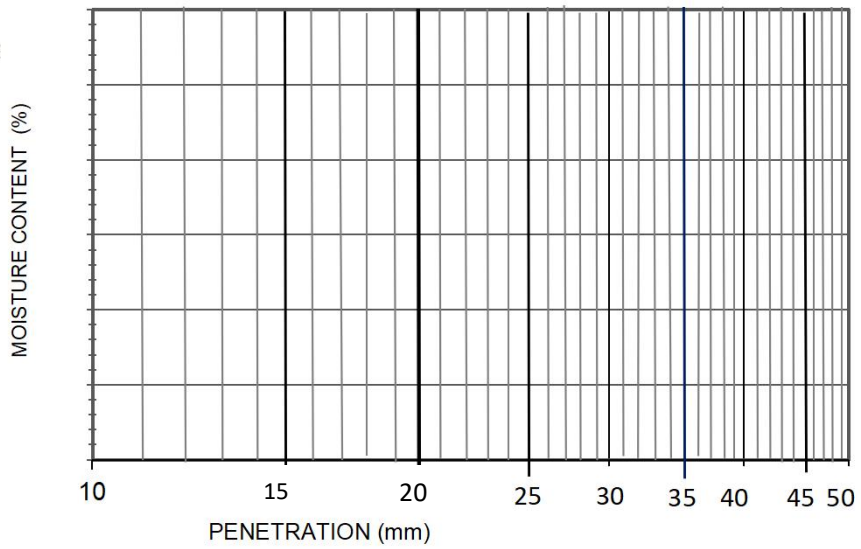
Pit No.: ----------

Depth: ----------

Sample History: Natural state ☐ Air dried ☐ Unknown ☐

Sample Preparation: Wet sieved ☐ Dry sieved ☐

| Ideal targeted penetration (mm) [1] | | | 10-15 | 16-20 | 21-25 | 26-30 | | | |
|---|---|---|---|---|---|---|---|---|---|
| Initial dial reading | mm | (R1) | | | | | | | |
| Final Dial Reading | mm | (R2) | | | | | | | |
| Cone Penetration depth | mm | ($R_2$ - $R_1$) | | | | | P.L | P.L | N.M.C |
| Container No. | | | | | | | | | |
| Mass of wet soil & container | g | M2 | | | | | | | |
| Mass of dry soil and container | g | M3 | | | | | | | |
| Mass of dry soil & container (check) | g | Mc | | | | | | | |
| Mass of container | g | M1 | | | | | | | |
| Mass of water (M2 - M3) | g | M4 | | | | | | | |
| Mass of dry soil (M3 - M1) | g | M5 | | | | | | | |
| Moisture content (M4 / M5 x 100) | % | **w** | | | | | | | |
| | | | | | | Aveage PL | | | |




**Linear shrinkage**

| | |
|---|---|
| shrinkage mould no. | |
| length of wet soil | |
| length of dry soil | |
| shrinkage | |
| shrinkage % | |

**SUMMARY**

| | | |
|---|---|---|
| N.M.C | % | |
| liquid limit @ pen 20 | wcL | |
| plastic limit | wp | |
| plasticity index | Icp | |
| linear shrinkage | LS % | |

**Material classifications / description**

**NOTE**: Check mass (Mc) must be used in all moisture calculations

Tested by: ---------- Checked by: ----------

Date: ---------- Date: ----------

Authorised by: AS (Material Science & Technology Branch)

**FORM No.S-3.9**

*Revised Worksheet: 08/04/2022 lhb*